

SecureGate: Mitigating Gate-Fingerprint Leakage in Fault-Tolerant Quantum Computing

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Circuit-level proof of concept for randomized virtual Pauli-error mitigation

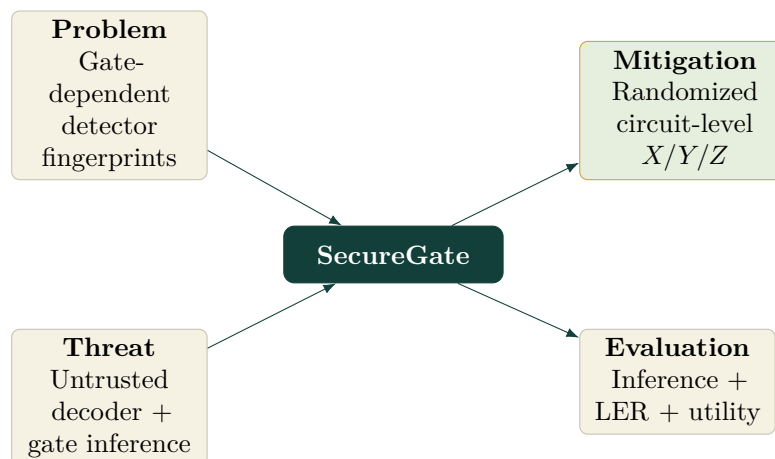
Abstract. SecureGate is a proof-of-concept security approach for fault-tolerant quantum computing that addresses decoder-side gate-fingerprint leakage. We study whether an untrusted decoder can infer hidden logical operations from spacetime syndrome data and evaluate randomized circuit-level virtual Pauli errors as a mitigation. In distance-5 surface-code simulations, mitigation reduced gate-inference accuracy from 53.81% to 53.41% while maintaining approximately 99.42% decoder utility.

1. Problem and Research Question

Fault-tolerant quantum computers repeatedly extract syndrome information and pass detector records to a classical decoder. Recent research identifies a confidentiality threat in which gate-dependent statistical structure in spacetime syndrome data can expose information about a hidden logical operation to an untrusted decoder [1].

Research question: Can randomized virtual Pauli errors introduced at the circuit level perturb gate-dependent detector information while preserving measured decoder performance?

2. SecureGate Mind Map



3. What Is New in SecureGate?

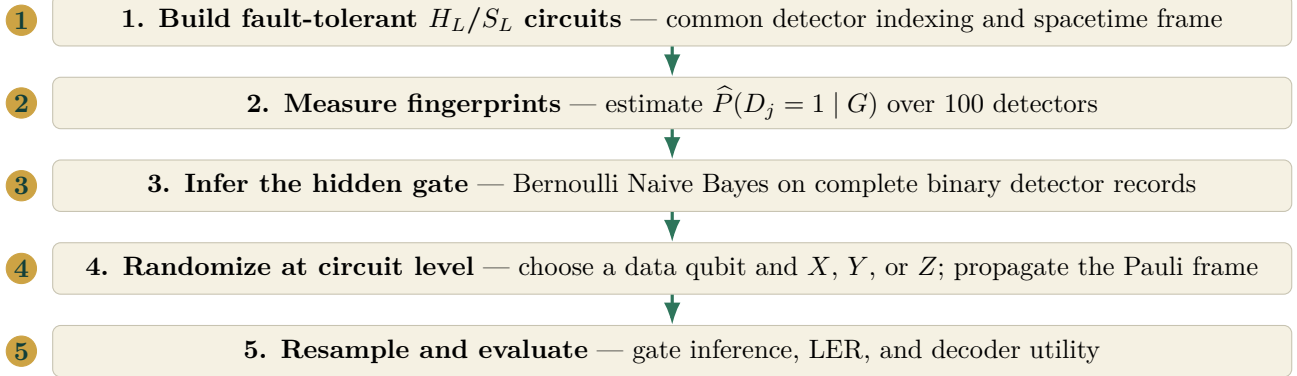
SecureGate connects gate-fingerprint measurement to an explicit circuit-level mitigation experiment. Rather than modifying detector records after measurement, the final experiment inserts a perfect Pauli instruction into the fault-tolerant circuit, propagates the Pauli frame, and directly resamples the altered circuit. The decoder used for evaluation is left unchanged. The Pauli-frame component follows the established principle of classical Pauli tracking through Clifford circuits [3].

Scientific scope. Gate-inference accuracy is an empirical measure of the performance of the specified Bernoulli Naive Bayes inference model. It is not a percentage of information leakage and does not quantify optimal adversarial inference.

4. Main Experiment at a Glance

Code / circuit	Unrotated CSS surface code, distance 5
Physical resources	81 physical qubits = 41 data + 40 ancilla
Logical operations	H_L and S_L ; 100 detectors; 1 logical observable each
Noise parameter	$p = 10^{-3}$
Sampling	10,000 shots per gate per run
Repeated evaluation	10 independent baseline runs + 10 mitigation runs
Mitigation space	41 data qubits $\times \{X, Y, Z\} = 123$ cases
Inference / decoder	Bernoulli Naive Bayes / matching decoder on S_L

5. Methodology



5.1 Detector Fingerprints

For detector j and logical gate G , the empirical event probability is

$$\hat{P}(D_j = 1 | G) = \frac{1}{N} \sum_{k=1}^N D_j^{(k)}, \quad N = 10,000.$$

Notebook 03 found mean detector-event probabilities of 0.024598 for H_L and 0.026675 for S_L . The mean absolute detector-wise difference was 0.003681, with a maximum of 0.019200 at detector 52 and $t = 1.0$. The full 100-detector profiles remained highly correlated ($r = 0.950529$) while retaining nonzero detector-level structure.

5.2 Gate Inference

A balanced dataset of 20,000 records (10,000 per gate) was evaluated with an 80/20 stratified split using Bernoulli Naive Bayes. The seed-42 test split gave 55.27% accuracy (95% Wilson interval [53.73%, 56.81%]). Repeated split, fresh-shot, and label-permutation controls supported a measurable but modest gate-dependent signal; the permutation control was 50.02%.

5.3 Circuit-Level Virtual Pauli Mitigation

The mitigation library contains all $41 \times 3 = 123$ data-qubit/Pauli cases. A virtual error is selected independently for each shot with probability $\alpha = 0.50$, with the data qubit and Pauli type sampled uniformly from the library. The same rule is used for H_L and S_L and does not depend on the hidden gate label.

A Pauli frame is represented by $I = (0, 0)$, $X = (1, 0)$, $Y = (1, 1)$, and $Z = (0, 1)$. For the relevant Clifford operations,

$$H : (x, z) \mapsto (z, x), \quad S : (x, z) \mapsto (x, z \oplus x),$$

with the corresponding controlled- Z and SWAP propagation rules. This follows the classical Pauli-tracking principle [3].

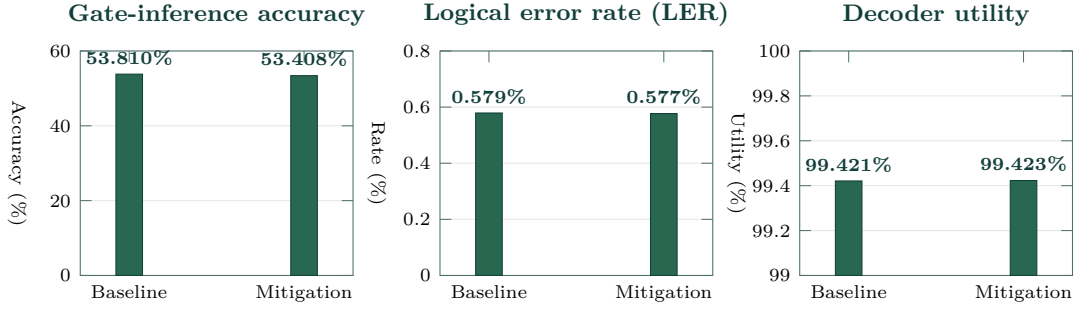
5.4 Evaluation

Gate-inference accuracy is the fraction of correctly classified test records. For the S_L decoder evaluation,

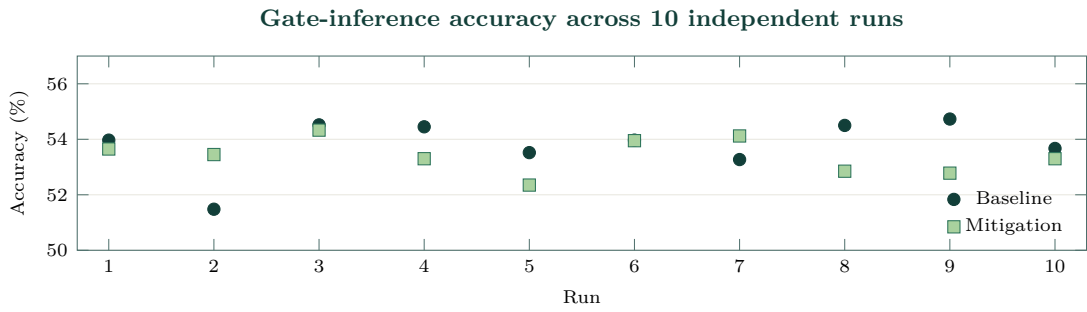
$$\text{LER} = \frac{N_{\text{logical errors}}}{N_{\text{shots}}}, \quad \text{Utility} = 1 - \text{LER}.$$

The final comparison uses ten independent runs per condition with fresh circuit sampling and new virtual-error realizations in the mitigation runs.

6. Results



Metric	Baseline	Mitigation	Change
Gate inference	53.810% $\pm$ 0.950 pp	53.408% $\pm$ 0.629 pp	-0.403 pp
LER	0.579% $\pm$ 0.087 pp	0.577% $\pm$ 0.104 pp	-0.002 pp
Decoder utility	99.421% $\pm$ 0.087 pp	99.423% $\pm$ 0.104 pp	+0.002 pp



Main finding. Across ten independent runs, gate-inference accuracy changed from 53.810% to 53.408% (about -0.403 percentage points), while LER and decoder utility remained essentially unchanged. The observed reduction is small and is not treated as statistically conclusive.

7. Interpretation

Leakage indicator. The baseline experiment shows a measurable, reproducible gate-dependent signal in the simulated H_L/S_L detector records, using Bernoulli Naive Bayes as a transparent baseline inference model.

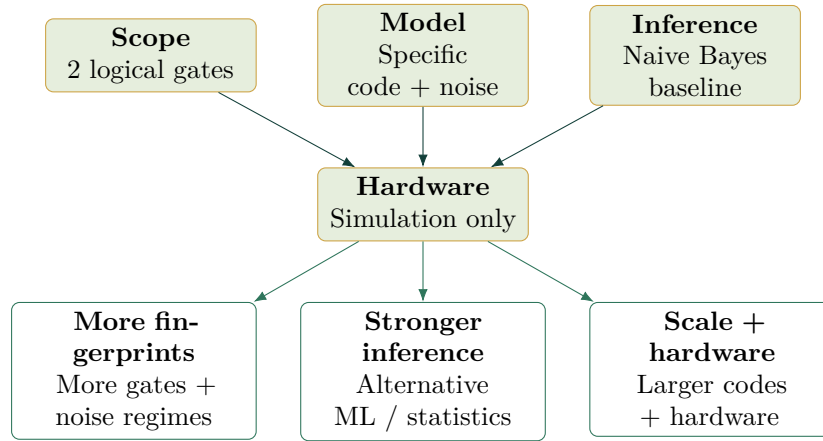
Mitigation effect. The circuit-level virtual-Pauli experiment is associated with a small decrease in measured gate-inference accuracy. The magnitude is modest relative to run-to-run variability, motivating broader evaluation.

Decoder performance. The measured LER changes by

-0.002 pp and decoder utility by $+0.002$ pp. Under the tested conditions, decoder performance therefore remains approximately unchanged.

Scope of the result. The findings provide a proof-of-concept evaluation under the tested simulation conditions. Further experiments with additional logical gates, stronger inference models, different noise regimes, larger code sizes, and real quantum hardware are needed to assess the generality of the approach.

8. Limitations and Future Work



9. Conclusion

SecureGate demonstrates a circuit-level proof of concept for evaluating randomized virtual Pauli errors as a mitigation strategy against decoder-side gate-fingerprint leakage. In the tested distance-5 simulation, gate-inference accuracy changed from 53.810% to 53.408%, while LERand decoder utility remained approximately 0.58% and 99.42%, respectively.

Overall finding. Circuit-level randomization was implemented and evaluated successfully, produced a small observed change in gate inference, and preserved comparable decoder performance under the tested conditions. The current evidence supports a proof-of-concept result and motivates broader experiments rather than a universal security claim.

10. Reproducibility and Public Resources

The project repository contains the Jupyter notebooks, source code, figures, circuit files, and supporting documentation:

<https://github.com/Marya6/Securegate-Saif>

The public report accompanies the reproducibility materials; the principal notebook chain is backend/state-preparation validation → gate interface → H_L/S_L fingerprint analysis → circuit-level mitigation.

References

- [1] S. Shukla, D. E. Browne, and S. Nishio, “Anticipating decoder side-channel attacks in fault-tolerant quantum computers,” arXiv:2607.12174, 2026.
- [2] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, “Surface codes: Towards practical large-scale quantum computation,” *Physical Review A*, vol. 86, 032324, 2012.
- [3] B. W. Reichardt, “Error-detection-based quantum fault tolerance against discrete Pauli noise,” UCB/EECS-2006-157, University of California, Berkeley, 2006.